

Continuous Monitoring Water Level Sensor

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I will develop a sensor with three improvements to the

Continuous Water-Level Sensor Based on Load Cell and Floating Pipe

research done by

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- Cost optimized
- Low power
- Environmentally friendly

Method to remotely monitor water levels for early warning indication of flooding.

Archimedes Principle

F_B = Weight Displaced

$$F_B = \rho_w g(V)$$

$$F_B = \rho_w g(\pi r^2 \cdot d)$$

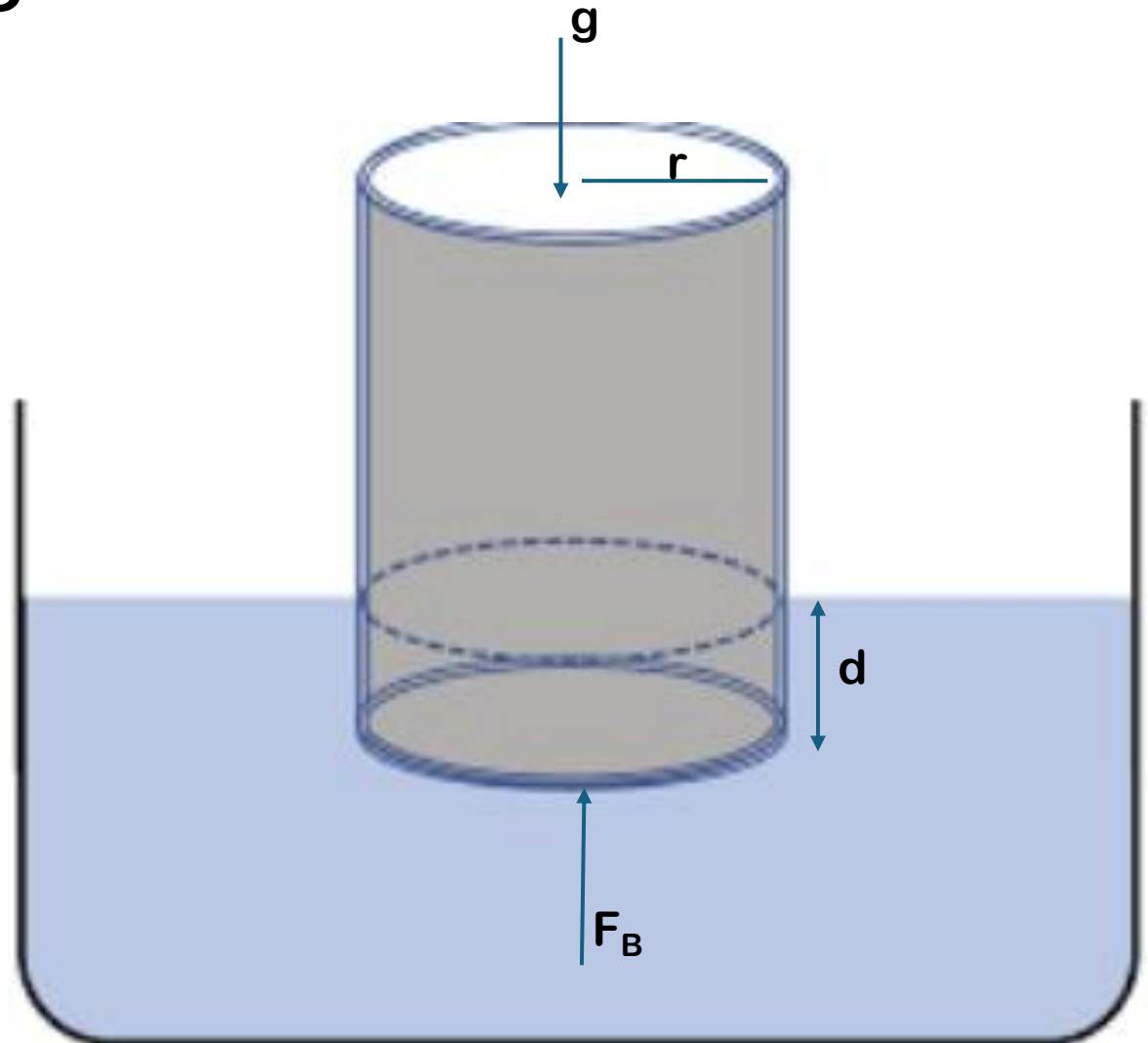
F_B = Buoyant Force

ρ = density of the water

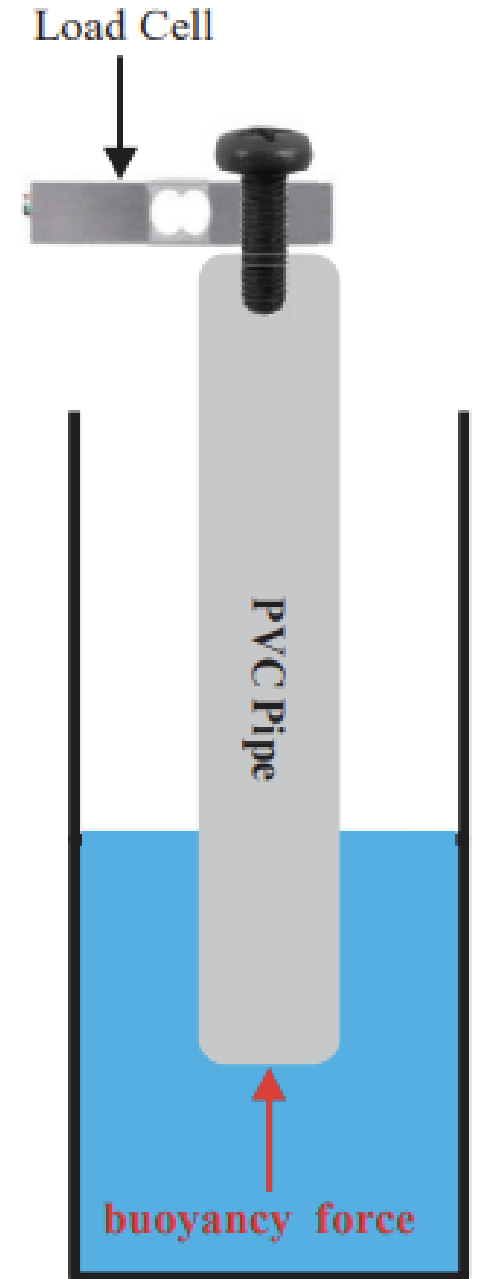
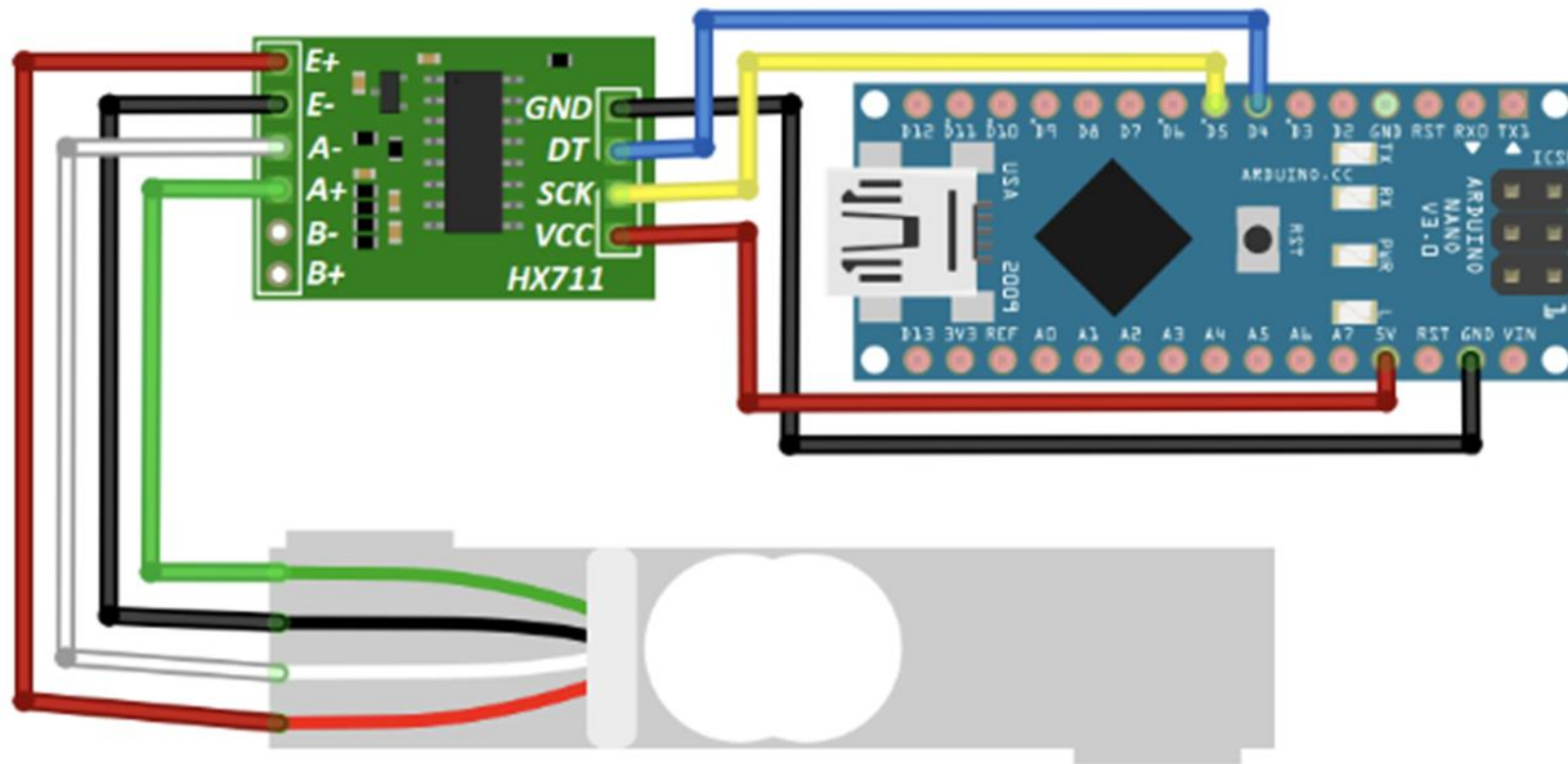
g = acceleration due to gravity

V = Volume of the displaced fluid

d = distance object is displaced



Floating Pipe and Load Cell Setup



LoRa WAN :

- Low Power, Wide Area (LPWA) networking protocol
- Wirelessly connect battery operated devices to network server via standard IP connections
- Bi-directional communication
- Converting RF packets to IP packets
- Uses **Long Range** RF

